

A Frobenius Group of Order 155: A Disproof of Conjecture 00000002305

GitHub: gaochengzhecpu

Abstract

The source asserts $k(G) \geq \frac{1}{2}[G : H]$ for the class number $k(G)$ of a finite group and a maximal subgroup H . The Frobenius group $G = C_{31} \rtimes C_5$ of order 155 has exactly 11 conjugacy classes and a maximal subgroup of index 31, and $11 < 31/2$. The group, the maximality of the subgroup, the exact class number and the negation of the universal bound are verified in Lean.

Reading of the source. Both language versions state the bound $k(G) \geq c[G : H]$ “for maximal H ” with $c = 1/2$, without restricting G . We read this as a claim about every finite group G and every maximal subgroup H of G , where $k(G)$ is the number of conjugacy classes of elements and a maximal subgroup is a proper subgroup contained in no other proper subgroup.

Theorem 1. *Let G be the group of the 155 affine maps $x \mapsto 2^k x + a$ of $\mathbb{Z}/31\mathbb{Z}$, where $a \in \mathbb{Z}/31\mathbb{Z}$ and $k \in \mathbb{Z}/5\mathbb{Z}$, and let $H = \{x \mapsto 2^k x\}$ be the stabiliser of 0. Then H is a maximal subgroup of index 31, and $k(G) = 11$. Consequently*

$$k(G) = 11 < \frac{31}{2} = \frac{1}{2}[G : H].$$

Proof. Since $2^5 = 32 \equiv 1 \pmod{31}$, the power 2^k depends only on k modulo 5, and 2 has multiplicative order exactly 5. Write (a, k) for the map $x \mapsto 2^k x + a$. Composition gives

$$(a, k)(b, l) = (a + 2^k b, k + l), \quad (a, k)^{-1} = (-2^{-k} a, -k),$$

so these maps form a group. Distinct pairs give distinct maps, because the values at 0 and 1 determine a and 2^k . Hence $|G| = 31 \cdot 5 = 155$.

Maximality. The subgroup $H = \{(0, k)\}$ has order 5 and index 31. Let $H < K \leq G$. By Lagrange’s theorem 5 divides $|K|$ and $|K|$ divides $155 = 5 \cdot 31$, so $|K| \in \{5, 155\}$. As $K \neq H$ and $H \leq K$, we get $|K| = 155$ and $K = G$.

Conjugacy classes. A direct computation gives, for $c = (b, l)$ and $g = (a, k)$,

$$c g c^{-1} = (2^l a + (1 - 2^k)b, k). \tag{1}$$

If $k \neq 0$, then $2^k \in \{2, 4, 8, 16\}$ and $1 - 2^k \in \{-1, -3, -7, -15\}$ is a unit modulo the prime 31. Taking $l = 0$ in (1), the first coordinate runs over all of $\mathbb{Z}/31\mathbb{Z}$ as b varies. Thus $\{(a, k) : a \in \mathbb{Z}/31\mathbb{Z}\}$ is a single class for each of the four values $k \neq 0$.

If $k = 0$, then (1) reads $c(a, 0)c^{-1} = (2^l a, 0)$. The class of $(a, 0)$ is therefore the orbit of a under multiplication by the subgroup $\langle 2 \rangle$ of order 5. The orbit of 0 is $\{0\}$. The 30 nonzero residues split into $30/5 = 6$ orbits. They are represented by 3^j , $0 \leq j \leq 5$, since 3 is a primitive root modulo 31.

In total there are $4 + 1 + 6 = 11$ classes. As a check, their sizes add up to $4 \cdot 31 + 1 + 6 \cdot 5 = 155$. □

Corollary 1. *The inequality $k(G) \geq c[G : H]$ fails for this pair for every constant $c > 11/31$. In particular it fails for $c = \frac{1}{2}$ and for $c = \frac{1}{2}(1 - \varepsilon)$ whenever $\varepsilon < 9/31$.*

Remark. The example belongs to a family. Let p be a prime and q a divisor of $p - 1$. The same argument shows that $C_p \rtimes C_q \leq \text{AGL}(1, p)$ has a maximal subgroup C_q of index p and exactly $q + (p - 1)/q$ classes. The ratio $q/p + (p - 1)/(pq)$ is not bounded below by a positive constant: for each q , Dirichlet's theorem gives a prime $p \equiv 1 \pmod{q}$ with $p > q^2$, and then the ratio is below $2/q$. For $q = p - 1$ one obtains $\text{AGL}(1, p)$ itself, with ratio 1. This remark is not used in the disproof and is not formalised.

Formal verification and scope

The Lean type `G` has two fields, $a \in \text{ZMod } 31$ and $k \in \text{ZMod } 5$, with the multiplication displayed above. The group axioms are proved, not assumed. The theorems `act_mul` and `act_faithful` show that `G` acts faithfully on `ZMod 31` by the affine maps $x \mapsto 2^k x + a$, so it is the group of the theorem. `card_G` gives the order 155.

`H` is a Mathlib `Subgroup`; `mem_H` identifies it with the stabiliser of 0. `index_H` proves `H.index = 31` for Mathlib's subgroup index. `H_maximal` proves `IsCoatom H`: H is a coatom of the lattice of all subgroups, which is the definition of a maximal subgroup.

The class number is the cardinality of Mathlib's quotient type `ConjClasses G`. `exists_rep` proves that every element is conjugate to one of eleven explicit representatives. The class function $(a, k) \mapsto (k, a^5 \text{ if } k = 0)$ is invariant under conjugation (`label_conj`) and separates the representatives (`label_rep`). Hence `card_conjClasses` proves that the number of classes is exactly 11; it is not an upper bound only.

The proposition `ClaimedBound` quantifies over every finite group and every coatom M of its subgroup lattice and states $\frac{1}{2} [\Gamma : M] \leq k(\Gamma)$ in $\mathbb{Q}$. The final theorem `conjecture_false` is its negation. `fails_above` is the corollary. Finite checks use kernel evaluation by `decide`; no `native_decide`, `sorry` or custom axiom occurs.

This refutes the bound as stated. The source does not define $\varepsilon(G)$, so the parenthetical variant has no precise statement beyond the corollary. Besides the conjugates of H , the only maximal subgroup of G is the normal subgroup of order 31; it has index 5 and satisfies the inequality. The example therefore does not address a reading in which H is only some maximal subgroup. No claim is made about the tightness clause.

Source and reproduction

The exact bilingual source is included byte-for-byte as `SOURCE.md`. Its repository path is

`conjectures/00000002305.md`.

The project pins Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From `lean/`, run `lake build`, then

`lake env lean -DwarningAsError=true Main.lean`.

The included public-Git manifest locks all dependency revisions. The `verification/` directory records the fresh-build logs, theorem-axiom reports, artifact hashes, review and final PDF inspection. No supplementary numerical computation is required.